

MOVIE RECOMMENDATION SYSTEM

Problem Statement

With the rapid growth of digital entertainment platforms and the vast availability of movies, users often face difficulty in discovering movies that match their personal preferences. Traditional search methods may not effectively provide personalized suggestions, leading to reduced user satisfaction.

This project focuses on developing a machine learning-based Movie Recommendation System that provides personalized movie recommendations based on user preferences and movie similarity. By analyzing user interactions and rating patterns, the system helps users discover relevant movies and enhances their viewing experience.

Objectives:

- Build a personalized movie recommendation system
- Analyze user preferences and movie similarities
- Recommend movies similar to user-selected titles
- Improve user experience through intelligent recommendations
- Develop a scalable recommendation solution for entertainment platforms

Dataset Used

Dataset Name:

MovieLens Dataset

Dataset Overview:

MovieLens is a widely used public dataset for recommender system research, containing:

- User ratings
- Movie metadata
- Movie identifiers
- User-item interactions

Dataset Features:

- User IDs

- Movie IDs
- Movie titles
- Ratings
- Movie metadata

Data Characteristics:

- Structured dataset
- Recommendation system problem
- Collaborative filtering task
- Large-scale user-movie interaction data

Approach / Methodology

Data Preparation:

- Loaded MovieLens dataset
- Preprocessed movie ratings and metadata
- Created a user-item matrix
- Converted data into sparse matrix format using CSR (Compressed Sparse Row)
- Built final dataset for similarity analysis

Recommendation Technique Used:

Collaborative Filtering:

Collaborative filtering recommends movies based on patterns of user behavior and similarities between items.

Item-Based Recommendation:

- Focuses on movie-to-movie similarity
- Uses previously liked movies to suggest similar titles

Feature Similarity Method:

Cosine Similarity:

Cosine similarity measures the similarity between movie vectors based on user ratings.

Formula:

- Smaller cosine distance indicates higher similarity

- Similar movies are recommended based on nearest neighbors

Machine Learning Model Used:

K-Nearest Neighbors (KNN):

KNN was implemented using:

- Algorithm: Brute force
- Metric: Cosine similarity

Workflow:

1. Search for input movie title
2. Retrieve movie ID
3. Locate movie index in dataset
4. Compute nearest neighbors
5. Sort similar movies by distance
6. Return top recommendations

Output

Movie Recommendation System

Enter a movie name to get recommendations:

Movie Name:

Recommended Movies:

	Title	Distance
1	Men in Black (a.k.a. MIB) (1997)	0.4499
2	Star Wars: Episode I - The Phantom Menace (1999)	0.4691
3	Saving Private Ryan (1998)	0.4704
4	Shrek (2001)	0.4812
5	Sixth Sense, The (1999)	0.486

Conclusion

This project successfully developed a machine learning-based Movie Recommendation System using collaborative filtering, cosine similarity, and K-Nearest Neighbors.

Future Scope

- Hybrid recommendation systems
- Content-based filtering integration
- Deep learning recommendation models
- Real-time streaming platform deployment
- User feedback adaptation
- Enhanced recommendation diversity

Business Impact

This system can help:

- Streaming platforms improve engagement
- Increase user satisfaction
- Enhance content discovery
- Improve retention rates
- Personalize user experiences

Applications

- Streaming services
- Movie review websites
- Personal movie libraries
- OTT platforms
- E-commerce entertainment systems